

From prediction to prescription

A stroke risk analytics pipeline



Most stroke models tell you who is at risk. This one asks the harder question — for each patient, which intervention lowers their risk most, and by how much?

The problem

Prediction alone doesn't prevent strokes — action does.

Stroke is the second-leading cause of death worldwide and the leading cause of adult disability. An estimated 80% of strokes are preventable through blood pressure control, glucose management, weight and smoking changes.

Yet most published stroke risk models stop at a probability score — a number, not a plan. For a clinician or a patient, that's the least useful part of the answer.

80%

of strokes are preventable

Source: World Stroke Organization / Lancet Neurology INTERSTROKE study

The dataset at a glance

Kaggle Healthcare Stroke Dataset · a real-world class-imbalance problem.

5,110

patients

12

features

4.9%

positive class rate

0

duplicate rows

Features captured

- Demographics: age, gender, ever married, work type, residence type
- Clinical: hypertension, heart disease, average glucose, BMI
- Behavioural: smoking status
- Target: stroke (binary)

Cleaning

- 201 missing BMI values imputed by (age × gender) group mean, with a gender-only fallback for empty buckets.
- One 'Other' gender row dropped for modelling — n=1 not learnable.
- Categorical variables one-hot encoded; binaries mapped 0/1; features standardised for linear and NB models.

A three-layer analytics stack

Descriptive → Predictive → Prescriptive · each layer answers a different question.

01**DESCRIPTIVE**

Who does this happen to?

Profile the sample and the stroke cohort. Distributions, comorbidities, age and glucose skews. Build the intuition a model has to respect.

EDA & data cleaning**02****PREDICTIVE**

Who is likely to be next?

Four base models and three ensembles benchmarked head-to-head under class imbalance. Full diagnostic pack: ROC, PR, calibration, SHAP.

Logistic · XGBoost · RF · NB · Stacking**03****PRESCRIPTIVE**

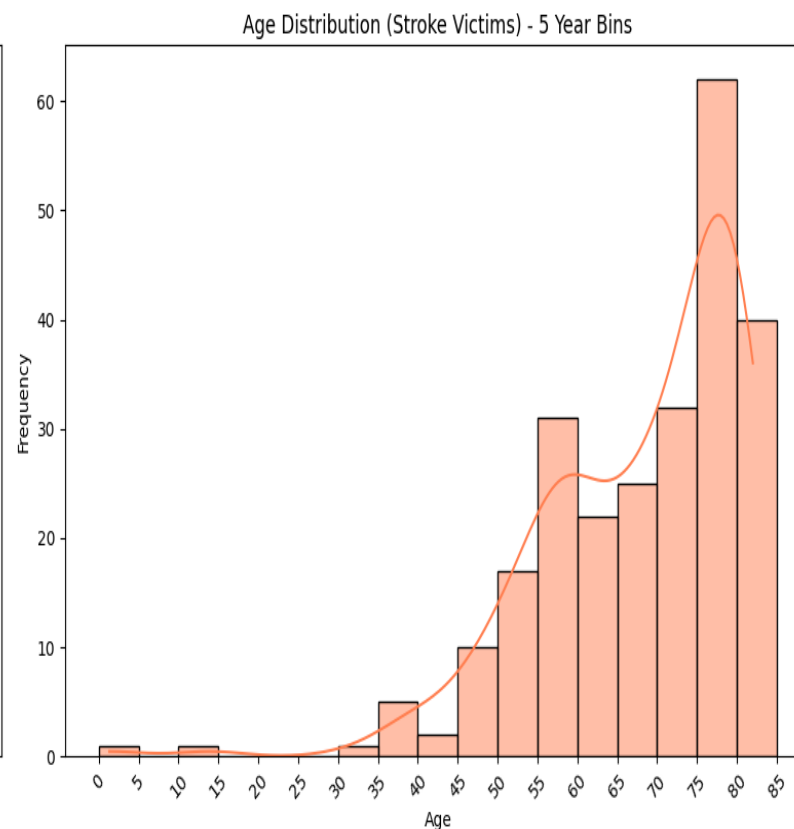
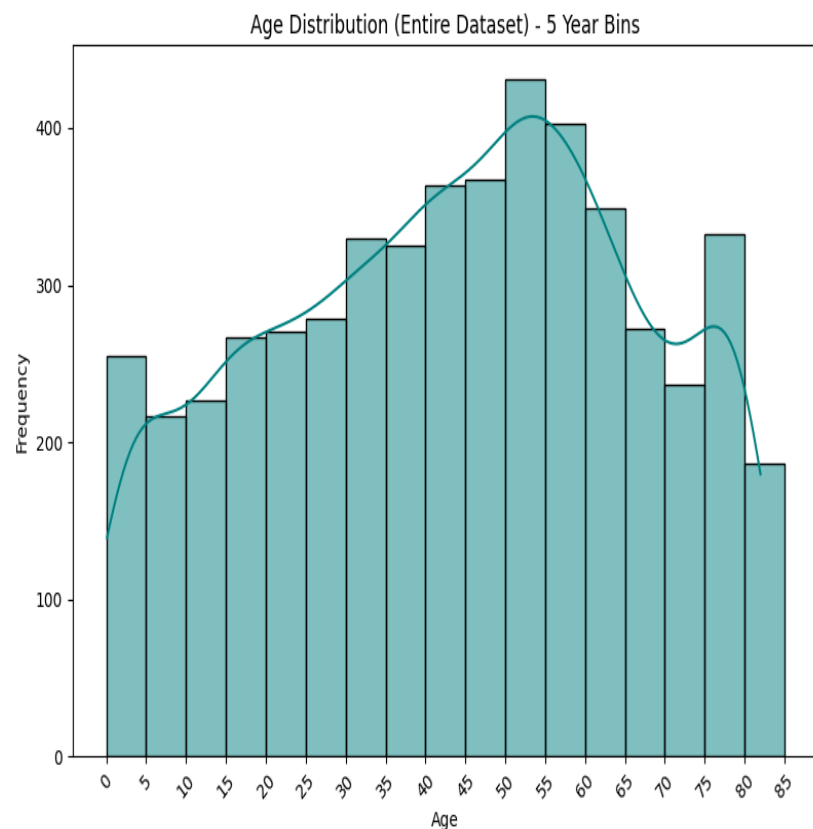
What should we do about it?

Calibrate probabilities so they mean what they say, then run per-patient counterfactuals: how much does risk drop if hypertension, glucose, BMI or smoking is addressed?

Calibrated risk % + counterfactual advice

Age dominates the signal

Stroke incidence rises sharply from age 55 and concentrates 65+.

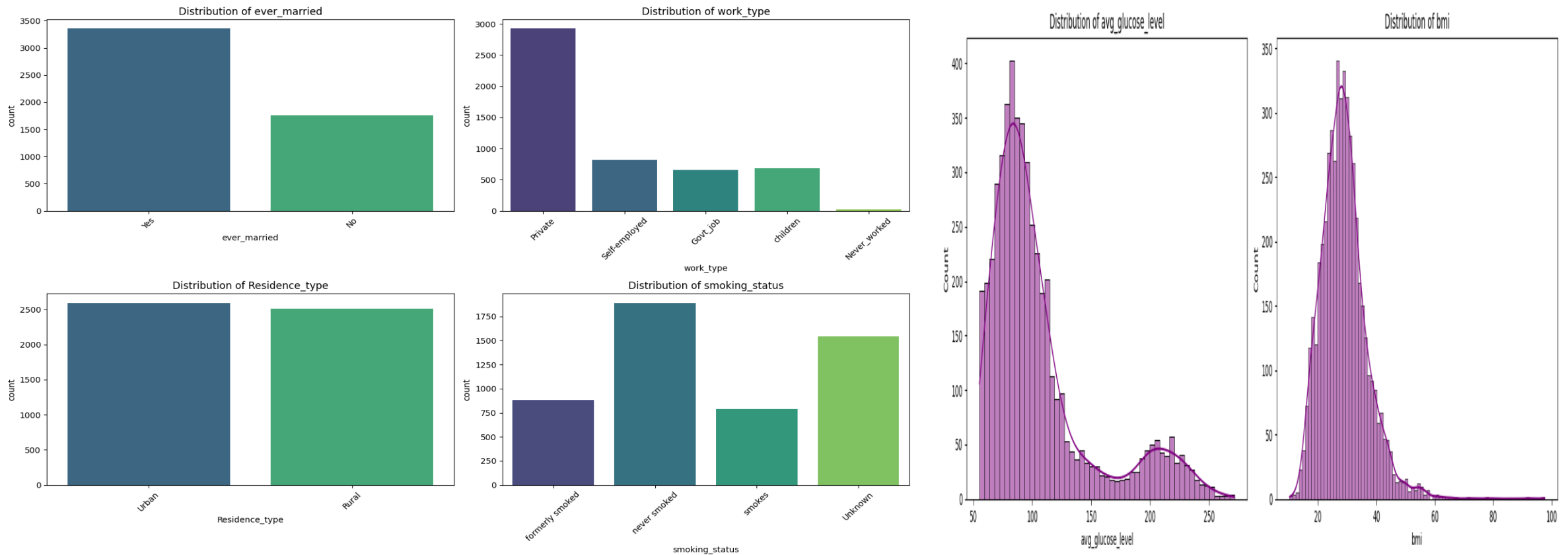


Key take-aways

- Whole-population age distribution is roughly uniform 0–80.
- Stroke cohort skews heavily into the 60–80 bracket.
- One outlier: a 1.32-year-old case — real-world data isn't clean.
- Age becomes the single strongest predictor in every model that follows.

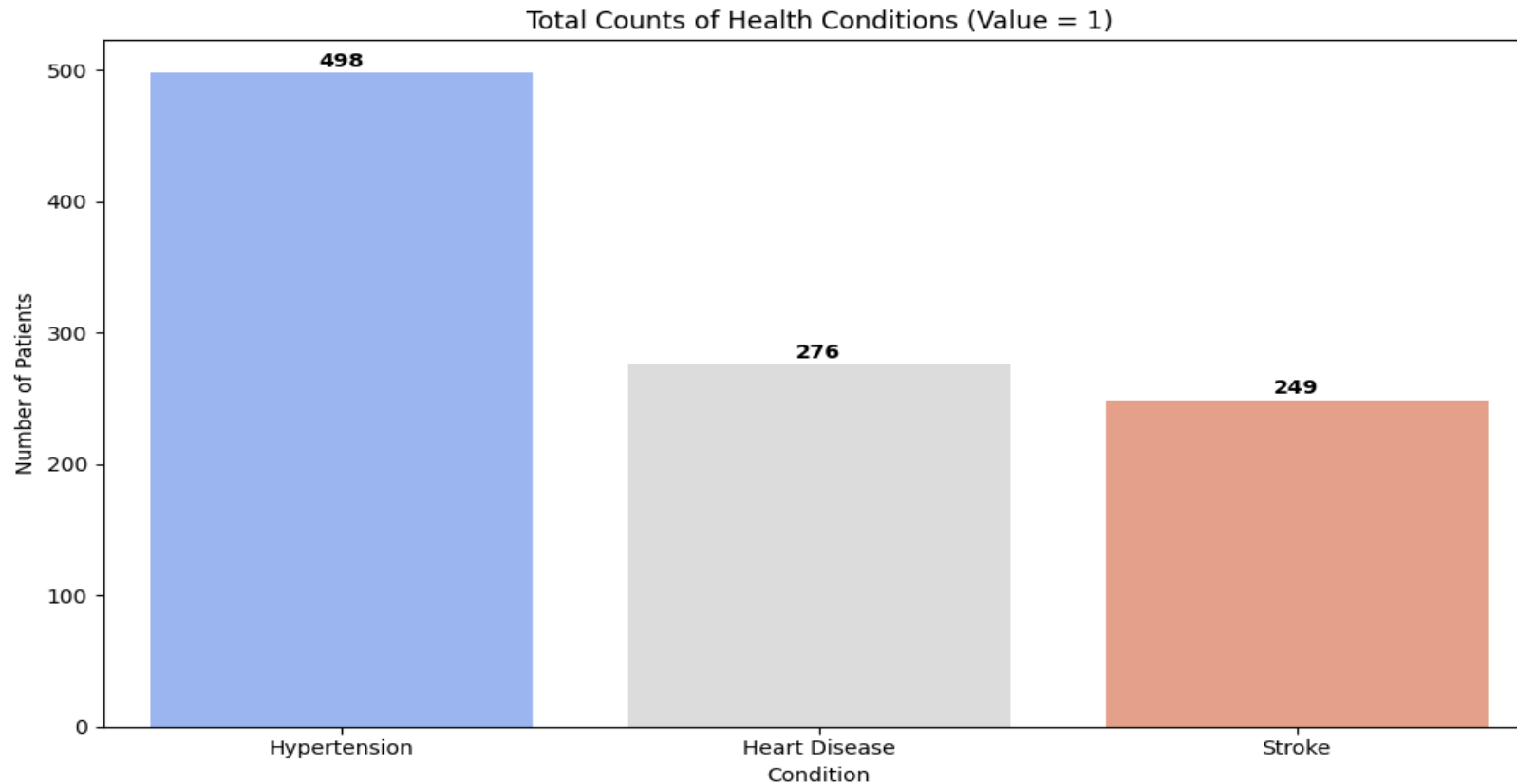
Feature distributions across the sample

Where the population sits on the modifiable risk factors.



Comorbidity totals in the sample

Hypertension and heart disease sit far above the stroke base rate.



Why this matters

- Stroke is a low base-rate event — ~5% of the sample.
- Comorbidities are ~10× more common than the outcome.
- Modelling has to handle class imbalance without letting the majority class dominate.
- Approach: `class_weight='balanced'` for linear / RF / NB, and `scale_pos_weight` for XGBoost.

Four models, three ensembles

Each model brings a different worldview — combined they cover the space.

Interpretable

Logistic Regression

Interpretable baseline. Standardised coefficients read like factor loadings — direct effect on log-odds of stroke.

Non-linear

XGBoost

Gradient-boosted trees. Handles non-linearities and interactions; SHAP values expose exactly what it learned.

Ensemble native

Random Forest

500 trees voting. Feature importance with std bars, plus partial-dependence plots that reveal each feature's functional form.

Probabilistic

Naive Bayes

Fast probabilistic baseline. Its Gaussian assumption is exposed by overlaying assumed vs. actual distributions.

Ensembles

Voting (equal & AUC-weighted)

Soft-vote averages of the four base models — equal weights vs. weighted by individual AUC.

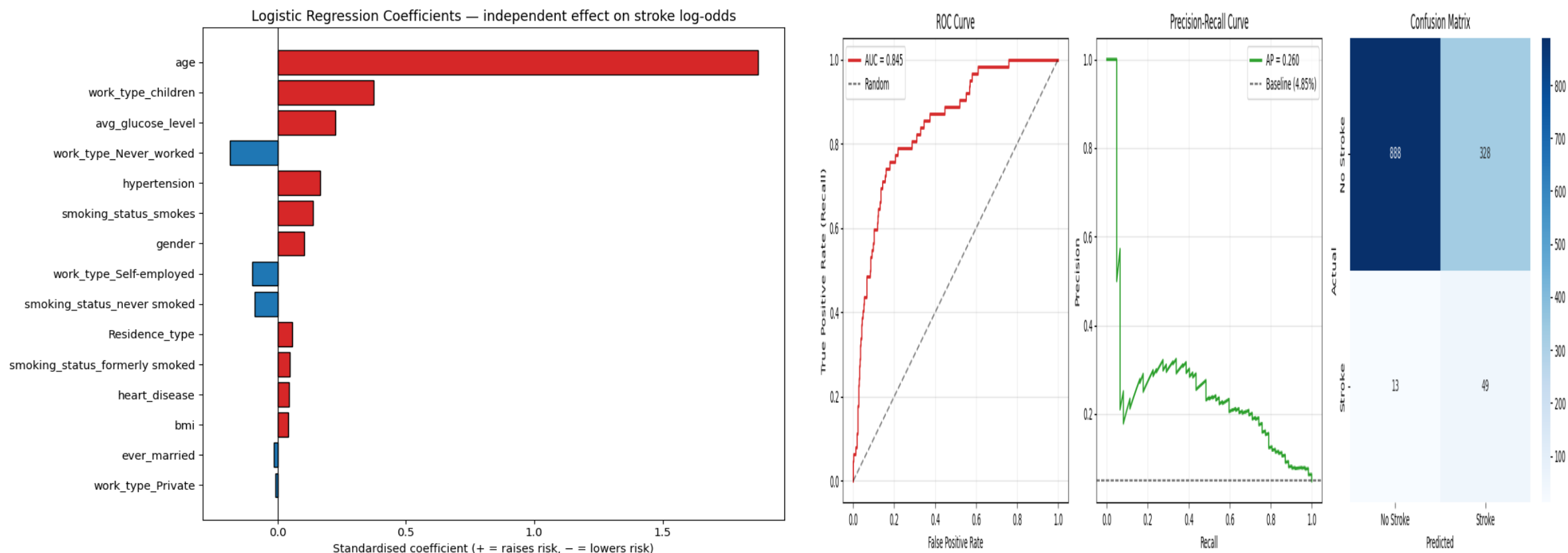
Meta-learner

Stacking

Meta-logistic regression trained on cross-validated base-model predictions — learns which model to trust when.

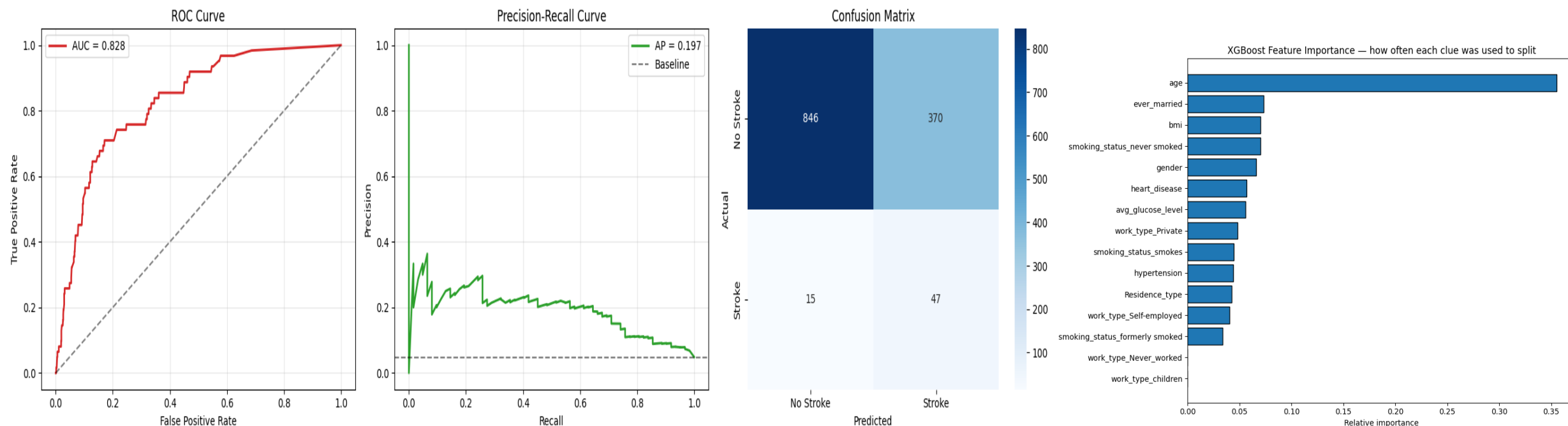
Logistic regression — the interpretable baseline

Coefficients read like standardised factor loadings — direct effect on the log-odds of stroke.



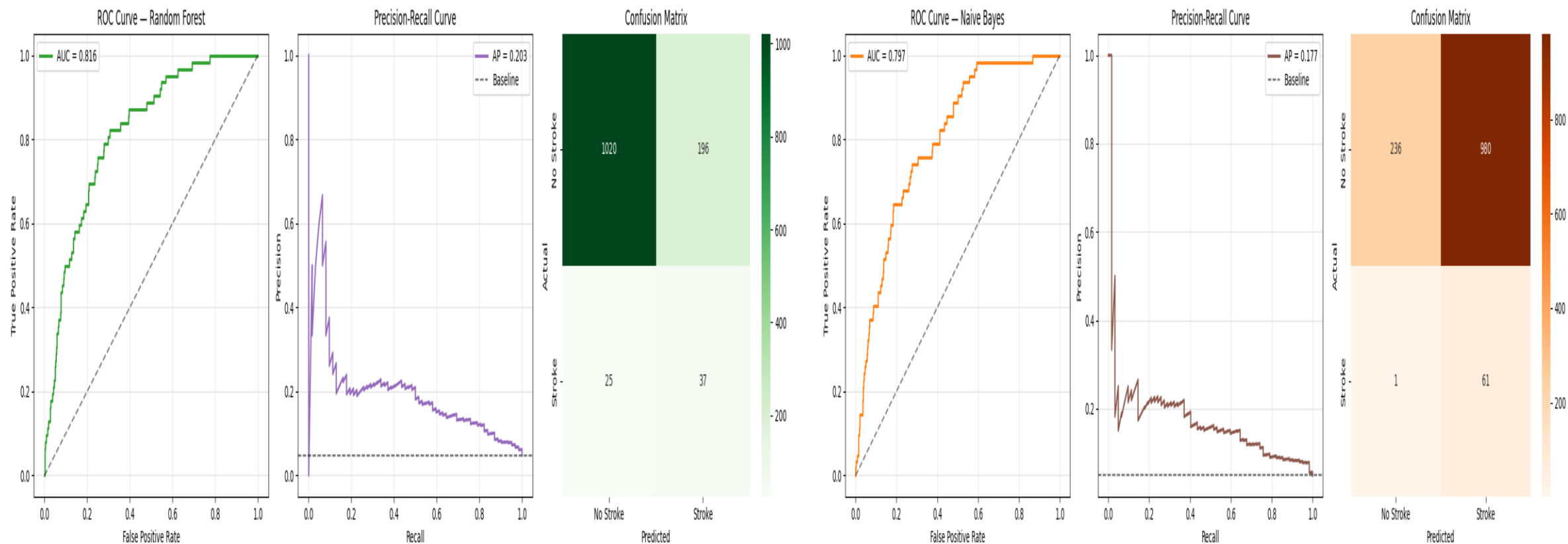
XGBoost — capturing non-linearities

SHAP values expose which features drove each individual prediction.



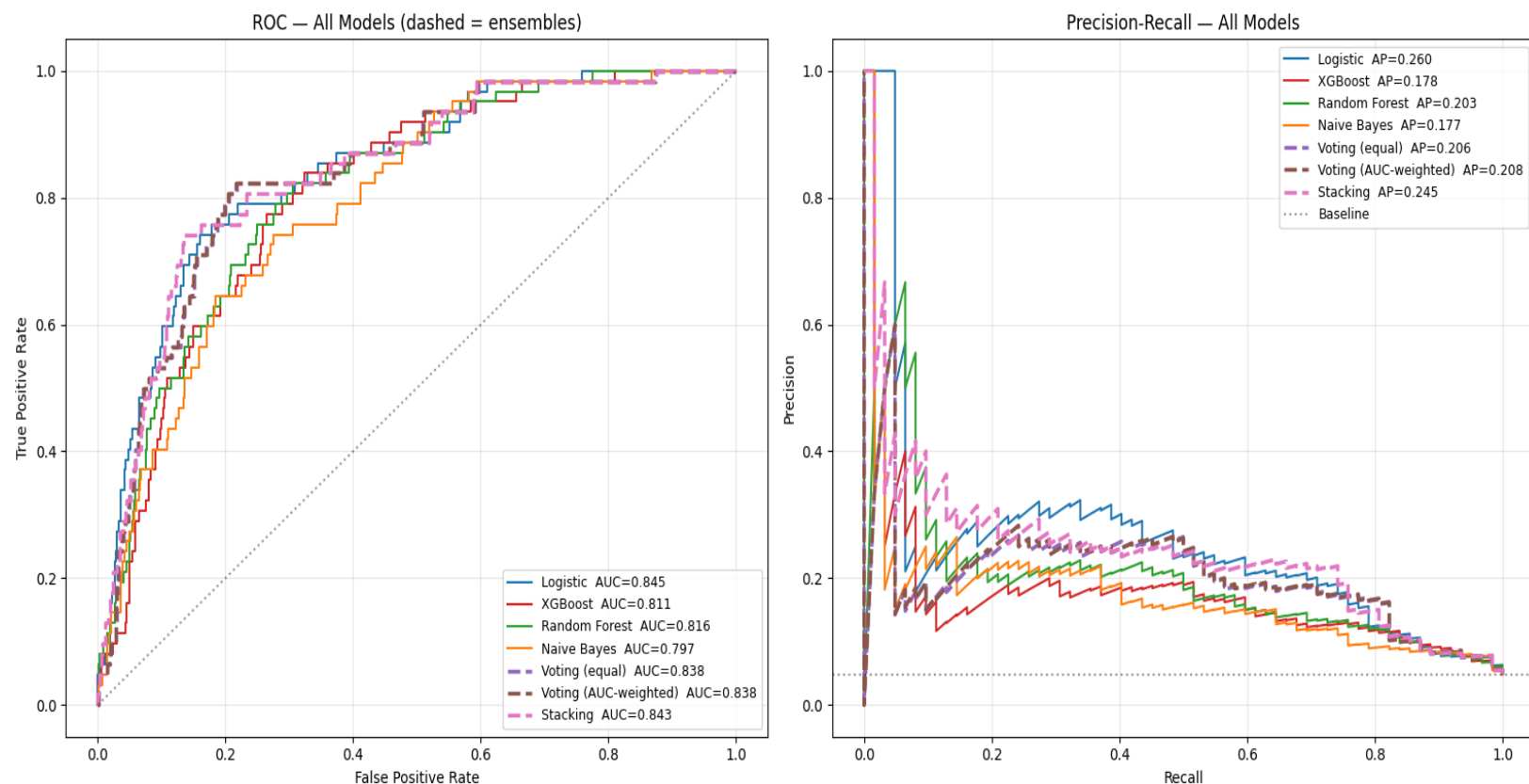
Random Forest & Naive Bayes

One bagged-tree learner and one probabilistic baseline — for coverage of the model space.



The leaderboard — ensembles vs. base models

ROC-AUC, PR-AUC and calibration compared head-to-head.

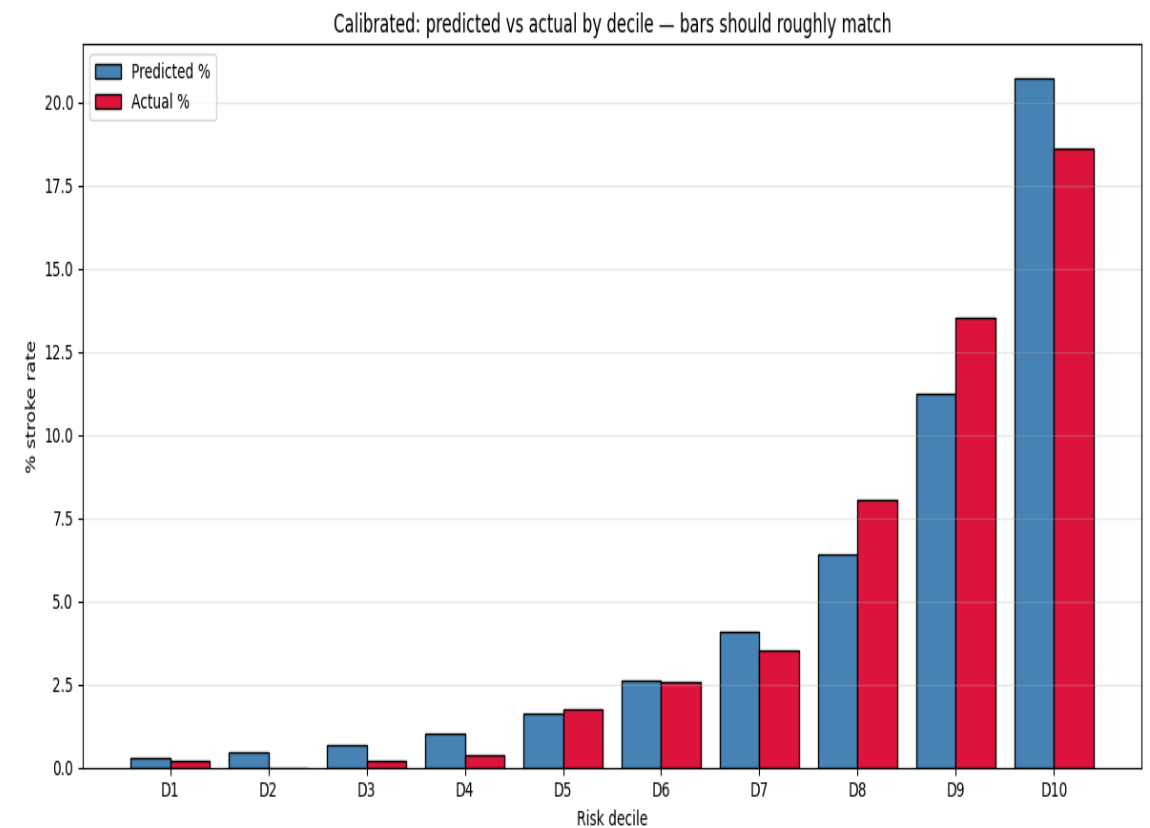
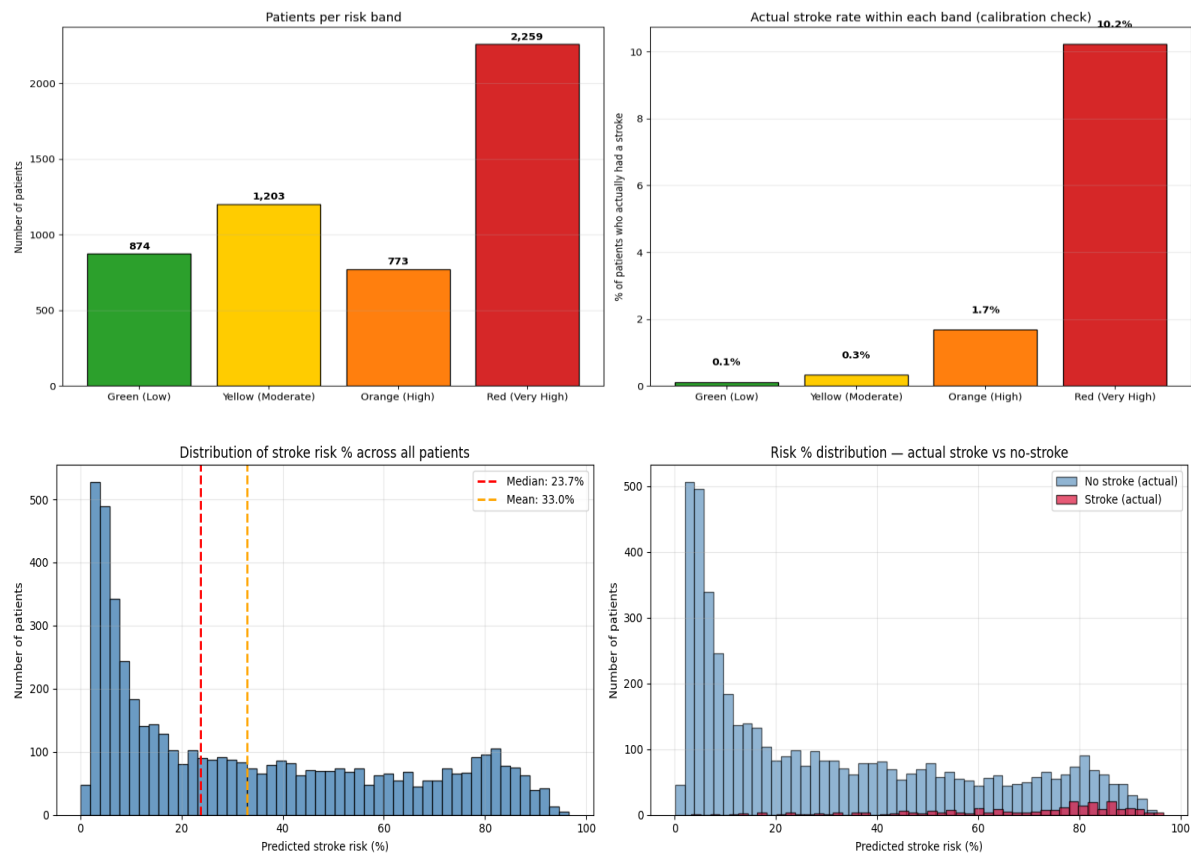


Reading the chart

- AUC alone is a misleading metric under 5% base rate — PR-AUC and calibration matter as much.
- The stacking ensemble edges base models on ranking, but gains are marginal.
- Naive Bayes' probabilities are systematically over-confident — visible in the calibration curve.
- Chosen for prescriptive layer: calibrated logistic regression — simple, honest, best calibration.

Calibrated risk % — bands that mean what they say

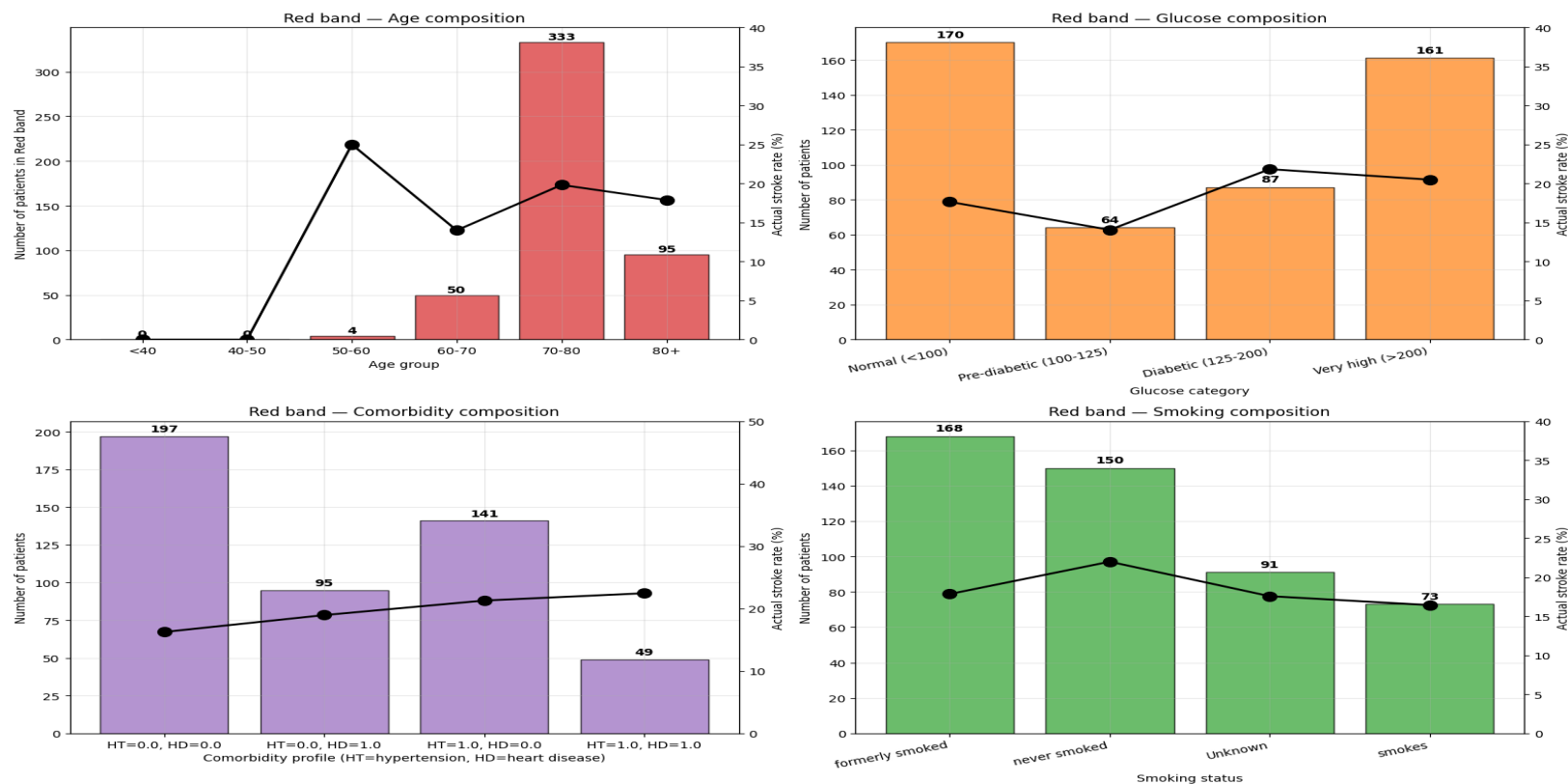
Platt scaling via 5-fold CV corrects class-weighted probabilities back to reality.



Anatomy of the Red band

Who ends up flagged Very High Risk — and are those flags clinically credible?

Red Band Clinical Anatomy — who ends up flagged as Very High Risk

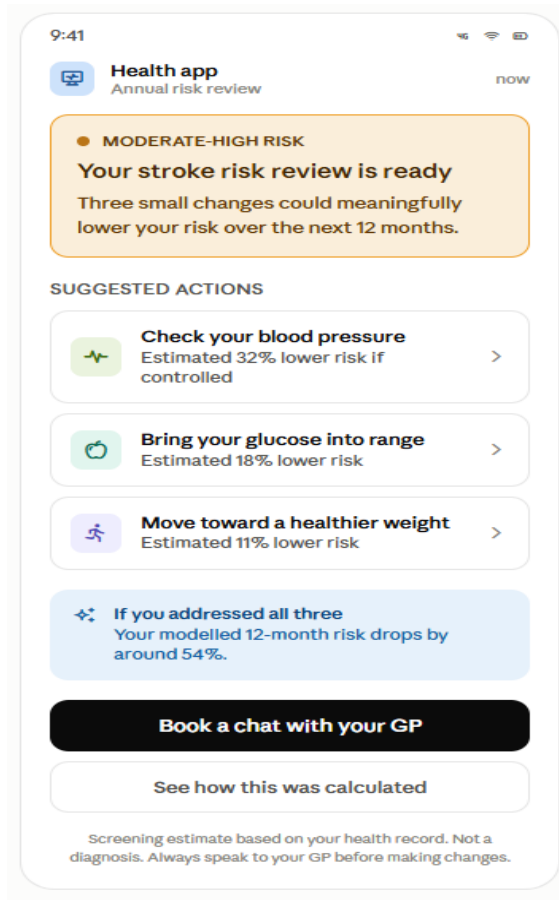


Clinical sanity check

- Red band concentrates the expected profile: older, hypertensive, elevated glucose.
- Actual stroke rate rises monotonically with model-predicted band.
- Counterfactual layer isolates the modifiable slice of risk — what a clinician can actually change.
- Each patient gets a natural-language advice string with ranked interventions.

The solution — a patient-facing risk alert

Where the analytics lands: an app alert built on the calibrated risk model and counterfactual advice.



Design principles

Language over numbers

A raw 22% figure without context creates anxiety, not action. The band and framing lead — the model's math sits behind a 'See how this was calculated' tap.

Actions ranked by modelled impact

Each intervention shows the estimated relative risk reduction — direct from the counterfactual layer in the notebook.

Route the patient to a clinician

Primary CTA is 'Book a chat with your GP'. Never a self-diagnosis loop.

Honest disclaimer

'Screening estimate. Not a diagnosis.' — the line that keeps the tool on the right side of MHRA / CE-marking rules.

Limitations & tools used

Limitations

- Screening tool, not a diagnostic. With a ~5% base rate, high recall costs precision at every threshold.
- Observational data. Counterfactuals are model-implied risk reductions, not causal estimates.
- Feature set is limited — no BP number, cholesterol, medication history or imaging.
- Single dataset, no external validation. A production model needs a held-out cohort from a different source.
- Age dominates. Much of the 'high risk' flag is really 'old' — the counterfactual layer isolates what's actually modifiable.

Tools used

Language & environment

Python 3 · Jupyter / Google Colab

Data

pandas · numpy

Modelling

scikit-learn · xgboost · shap · CalibratedClassifierCV

Visualisation

matplotlib · seaborn

Export & delivery

xlsxwriter · README + this deck